

Performance Testing

Date	30 JUNE 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Web Browser
Type of Testing	Functional Testing
Target Module / API	Credit Card Approval Model
Test Environment	Local Machine(Flask Development Server)
Test Date	30-06-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Verify credit card approval prediction using valid applicant details.	1	10	The application successfully predicts Approved or Rejected based on the input data.
2	Submit multiple prediction requests consecutively to verify application stability.	5	30	The application processes all requests correctly without errors or crashes.
3	Test the application with invalid or incomplete input values.	1	15	The application displays validation messages or prevents submission until valid data is entered.

4	Verify response time for prediction requests under normal usage.	10	60	The application returns prediction results quickly (within a few seconds) and remains responsive.
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.85 seconds	Pass	Prediction generated quickly under normal usage.
2	Response Time (Max)	< 5 seconds	1.42 seconds	Pass	Maximum response time remained within the acceptable limit.
3	Throughput (Req/sec)	5 Req/sec	6Req/sec	Pass	Flask application handled multiple requests successfully in the local environment.
4	Error Rate	< 1%	0%	Pass	No errors occurred during functional testing.
5	CPU Utilization	< 80%	35%	Pass	CPU usage remained low during prediction.
6	Memory Utilization	< 80%	42%	Pass	Memory consumption remained within acceptable limits.

Step 4: Observations & Analysis

Key Findings:

- The Flask application successfully processed user inputs and generated credit card approval predictions with a quick response time.
- The XGBoost model achieved the highest performance among the evaluated models and was selected for deployment.
- The application correctly integrated the trained model, scaler, and label encoders, producing consistent prediction results.
- All implemented functional features were tested successfully in the local environment.

Bottlenecks Identified:

- The dataset was highly imbalanced, resulting in significantly more rejected applications than approved ones.
- Initial model deployment issues occurred due to loading an outdated model with a different number of input features.

- Encoding and feature mismatch between the training pipeline and Flask application caused incorrect predictions until the preprocessing pipeline was synchronized.

Optimization Steps Taken:

- Performed feature engineering by converting DAYS_BIRTH to AGE and DAYS_EMPLOYED to YEARS_EMPLOYED, improving feature representation.
- Removed unnecessary features and retained only the most important features based on feature importance analysis.
- Selected the XGBoost classifier as the final model because it achieved the best overall performance.
- Saved and integrated the correct trained model, scaler, and label encoders into the Flask application to ensure consistent preprocessing and accurate predictions.
- Improved the user interface by replacing encoded values with user-friendly dropdown options and enhancing the annual income input field for better usability.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below:

Credit Card Approval Prediction

Fill in the applicant details to predict credit card approval.

Gender	Own Car
Select Gender	Select
Own Realty	Annual Income
Select	Enter Annual Income
Income Type	Education
Select Income Type	Select Education
Family Status	Housing Type
Select Family Status	Select Housing Type
Occupation	Number of Family Members
Select Occupation	Enter Number of Family Members
Age	
Enter Age	
Years Employed	
Enter Years Employed	

Predict Credit Card Approval

Credit Card Approval Prediction

Fill in the applicant details to predict credit card approval.

Gender

Female

Own Car

No

Own Realty

Yes

Annual Income

216000

Income Type

Pensioner

Education

Secondary / Secondary Special

Family Status

Widow

Housing Type

House / Apartment

Occupation

Unknown

Number of Family Members

1

Age

54

Years Employed

4

Predict Credit Card Approval

Credit Card Approved 

Credit Card Approval Prediction

Fill in the applicant details to predict credit card approval.

Gender

Male

Own Car

No

Own Realty

Yes

Annual Income

200000

Income Type

Student

Education

Higher Education

Family Status

Single / Not Married

Housing Type

House / Apartment

Occupation

Unknown

Number of Family Members

3

Age

21

Years Employed

0

Predict Credit Card Approval

Credit Card Rejected ✖